

Attitude Stabilization of Quadruped Walking Robot

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Abstract-- This paper deals with estimating the attitude of a walking quadruped robot in the presence of engine noise. Contrary to wheeled robots, legged robots are inherently supposed to work in uneven environment. Therefore estimating the attitude of walking legged robot becomes the utmost important factor in designing such machines in order to control them. We have used IMU as the prime sensor for stabilized attitude estimation. In fact accelerometer is used for roll/pitch observations in stationary/quasi-stationary state. But this data is not suitable unless pre-processing. This involves the pre-filtering and error factor determination for raw IMU data. Engine vibration effects are removed by low-pass filter and error-factor are calculated by least square fit to raw IMU data. Finally results of applying these simple techniques show the stability of attitude estimation on quadruped walking robot in real experiment.

Index Terms— Attitude Estimation, IMU Calibration, Quadruped Robot, Engine vibration.

I. INTRODUCTION

Walking robots have proven to be a promising locomotion system, capable of performing tasks that conventional vehicles cannot perform. Based on the number of legs the robot has, there are bipeds like human or birds, quadruped like mammals and reptiles, hexapods like insects and octopods like spiders [1]. Among them, the quadruped walking robots imitate the large animals we interact with in our everyday life. It is known that less than about half the Earth's landmass is accessible to existing wheeled and tracked vehicles. But people and animals using their legs can go almost anywhere [2]. Legged robots can travel in outdoor terrain that is too steep, rutted, rocky, wet, muddy and snowy for conventional vehicles. Robots meeting such requirements must have terrain sensors, sophisticated computing and power systems, advanced actuators and dynamic controls. Quadruped walking machines are built to work in an obstructed and rough terrain. The most fundamental requirement for any mobile platform, whether autonomous or semi-autonomous, is state estimation where the states typically are position, velocity and orientation, called navigation data. States, especially posture or orientation estimation, is utmost important for walking robots in difficult uneven terrain where a sense of balance is absolutely necessary as it is the basis of

attitude control.

With attitude we refer to the robot's orientation relative to the gravity vector, usually described by pitch and roll angles. Contrary to wheeled or flying robots, legged walking robots have very unpredictable posture at times to work in uneven/rough environment, they are actually built for. Attitude sensors used in an application depend on the type of mobile robots and the working environment. But one commonly used attitude sensor is gyroscope and accelerometer.

Aided attitude estimation can be obtained by combining measurements from different kinds of sensors: rate gyros, inclinometers, accelerometers, multi-antennae GNSS. It is possible to use a rate gyro to derive attitudes by integrating the rigid body kinematic equations as well. With high-quality gyros and good initial values, these estimates may be accurate over long periods of time. However, due to mathematical integration process involved in obtaining attitude from rate-gyro, it becomes inevitable to aid this gyro-only attitude with other sensor using some fusion algorithm. It would also be desirable to use cheap gyros as high-quality gyros are very expensive. To provide an absolute reference of the attitude, the accelerometers derived attitude which relates the body to the gravity vector can be used. The accelerometer sensors also sense the translational/dynamic accelerations, so accelerometers can only be used for pitch/roll angle observation during low accelerations phase e.g stationary/quasi-stationary states.

Another unavoidable factor in achieving stabilized attitude of walking robot is the engine vibration effect on IMU data. Many researchers have studied this effect for different kind of platforms [3,4]. Walking robots body motion is inherently three dimensional, hence making most of its kinematics representations nonlinear. For nonlinear problems, Extended Kalman filters (EKF) are known to work successfully in many practical non-linear state estimation problems.

Attitude estimation via different combination of the above mentioned sensors has been studied by many authors such as Vaganay, Aldon, and Fournier[5], Barshan and Durrant-Whyte[6], Foxlin[7], Baerveldt and Klang [8]. Rehbindler and Hu.[9] designed an algorithm for fusing inclinometer and gyro data that could be shown to be stable and convergent. In legged robots, the attitude estimation problem alone was addressed in walking humanoid robots [10] and quadruped robots [11]. Many applications of quadruped robots are listed in [1].

In our work we are trying to achieve stabilized attitude estimation for quadruped walking robot in natural setting.

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Robot can have any kind of attitude while working/walking in obstructed uneven environment. For this, we aid gyro estimated attitude with accelerometer measurements of gravity during low speed as in our case. Before applying to the fusion algorithm, we apply pre-processing to IMU data.

Our paper consists of following section. After introduction in section-I, in section-II we will go after the theoretical and mathematical development of the underlying problem. Section-III deals with experimental setup and methodology. Obtained results are discussed in section-IV and we conclude our work in this section with some comments on future research work.

II. THEORETICAL AND MATHEMATICAL DEVELOPMENT

In this section, we will develop the necessary theoretical and mathematical background of the problem at hand i.e stabilized attitude estimation for quadruped walking robot. As mentioned in introduction part that for walking robots, full attitude especially stabilized roll/pitch are very important to be estimated accurately. This enables robot to walk smoothly on uneven terrain. Though heading angle information is not observed directly in our application, its errors are reduced through application of EKF.

A. Attitude Dynamics

Attitude provides information about an object's orientation with respect to the local level frame (horizontal plane) and true north. Attitude solution typically consists of three components: roll, pitch and yaw. Angular measurement on the horizontal plane, with respect to true north, is referred to as yaw. Yaw is also known as azimuth or heading. The angular measurement on the vertical plane, with respect to the local level frame, is computed as pitch or roll. Our attitude estimator is based on the minimal sensor observations and hence we tried to be self-contained i.e our attitude estimation algorithm does not rely on any external source which can be interrupted intentionally or unintentionally. Primary attitude sensor is IMU being self-contained and provide high data rate which is crucial for real-time control of robot.

There are three well-known methods to calculate attitude of a body containing gyro readings as input. These are:

- Euler method
- DCM method
- Quaternion

Here we will describe the Euler method being simple and conceptually appealing method. The Euler angles can represent the attitude of a platform from three successive rotations of the navigation frame to body frame. However the sequence of rotations is unique to achieve the given final attitude so the order of rotation should be defined first.

In this work, the navigation frame is rotated and fitted into the body frame in the sequences of yaw(ψ), pitch(θ), and roll(ϕ). Hence the transformation matrix from the navigation to the body frame, C_n^b , is constructed by

multiplying the consecutive rotation matrixes in the same sequences, that is,

$$C_n^b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\phi & s_\phi \\ 0 & -s_\phi & c_\phi \end{bmatrix} \begin{bmatrix} c_\theta & 0 & -s_\theta \\ 0 & 1 & 0 \\ s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} c_\psi & s_\psi & 0 \\ -s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Where $C_{(\cdot)}$ and $S_{(\cdot)}$ stand for $\sin(\cdot)$ and $\cosine(\cdot)$ respectively. Transformation from b-frame to n-frame is simply obtained by transpose of C_n^b i.e.

$$C_b^n = (C_n^b)^{-1} = \begin{bmatrix} c_\theta c_\psi & -c_\theta s_\psi + s_\phi c_\theta c_\psi & s_\phi s_\psi + c_\phi s_\theta c_\psi \\ c_\theta s_\psi & c_\phi c_\psi + s_\phi s_\theta c_\psi & -s_\phi c_\psi + c_\phi s_\theta s_\psi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix}$$

As the body frame rotates with respect to the navigation frame, the Euler angle also changes based on the gyro measurements of the angular rate vector. Hence the differential equation for the Euler angles should be constructed and solved to get Euler angles. Based on the sequence of rotations from the navigation to body frame, the angular rate of the body frame measured from the gyro can be transformed into the Euler angle rate. Since the rotation in the roll axis occurs last, the roll rate, $\dot{\phi}$, is equivalent to the angular rate along the x-axis of the body frame, ω_x . The pitch rate, $\dot{\theta}$, however, is transformed according to the roll angle, and the yaw rate, $\dot{\psi}$, is also transformed according to the roll and pitch angle. The resulting angular rates in the body frame are the sum of these three Euler rates, i.e:

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\phi & s_\phi \\ 0 & -s_\phi & c_\phi \end{bmatrix} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\phi & s_\phi \\ 0 & -s_\phi & c_\phi \end{bmatrix} \begin{bmatrix} c_\theta & 0 & -s_\theta \\ 0 & 1 & 0 \\ s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix}$$

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

The inverse of this equation gives an expression for the rates of Euler angles:

$$\dot{\Psi}(k) = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi / c_\theta & c_\phi / c_\theta \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

$$\dot{\Psi}(k) = [R] \omega_{ib}^b \quad (1)$$

where $[R]$ is the transformation matrix from the body-axes angular rates to the Euler angle rates and $\omega_{ib}^b = [\omega_x \ \omega_y \ \omega_z]^T$ is gyro output.

Equation (1) relates the body angular rates to the rates of Euler angles in a nonlinear form. Note that the solution

requires integration of the first-order nonlinear equation. The solution of equation (1) in discrete time can be obtained by using numerical integration, that is,

$$\begin{bmatrix} \phi(k) \\ \theta(k) \\ \psi(k) \end{bmatrix} = \begin{bmatrix} \phi(k-1) \\ \theta(k-1) \\ \psi(k-1) \end{bmatrix} + \int_{k-1}^k \begin{bmatrix} \dot{\phi}(k) \\ \dot{\theta}(k) \\ \dot{\psi}(k) \end{bmatrix} dT$$

$$\Psi(k) = \Psi(k-1) + \dot{\Psi}(k)dT \quad (2)$$

This provides attitude update.

B. Calibration and Alignment

Calibration is the process of determining the errors in IMU sensor and alignment is the process of determining the initial attitude for strapdown navigation.

The IMU sensor errors can be divided into deterministic and stochastic errors. Most deterministic errors, such as the fixed bias and scale factor error, can be identified in laboratory tests. One simple method is the six-position calibration test which exploits the measurement of the gravity and the Earth rotation rate, whose values are known to within a certain precision. Thus the bias and scale factor error of the IMU can be effectively identified [11]. However this method needs IMU to be able to rotate in different orientations, thus not suitable for strapped sensors. In this work, we have adopted a different scheme. We determine the errors using stationary IMU data and applying Least Square method. The gyro errors are determined by fitting least square linear curve to attitude data and determining the slope of the increasing attitude in stationary state. Ideally it should remain fixed but due to errors in gyro data, calculated attitude increase linearly. For acceleration error compensation, we first transform the acceleration in the navigation frame and calculate velocity. The velocity, for stationary data, in navigation frame exhibits almost linear profile in stationary state. Hence we can fit a linear curve in a same way as we did for attitude profiles. Thus we find error-factor for acceleration. By compensating the acceleration with this error-factor, we can use the resulting acceleration for calculation of velocity and then position in navigation frame. The whole schematic is shown in Fig. 1. Results of applying this calibration method are shown in Fig. 2 and Fig. 3. Fig. 2 shows the roll angle plot for static data and applying a least-square linear fit to this roll angle profile. After calculating the error-factor i.e slope of this least-square fit line, we removed this error factor from gyro readings and recalculated roll angle as shown in Fig.3. As expected, the roll angle remains near constant for stationary time period.

The next step we did using stationary data is IMU alignment. The initial roll and pitch angles are computed directly from accelerometer data averaged over the stationary period. The initial heading angle, however, is determined from the gyro measurements of the Earth's rotation rate in the static condition [12]. The gyro used in our work is of tactical grade, Honeywell's HG1700,

which has enough resolution to detect the Earth's rotation rate in stationary condition.

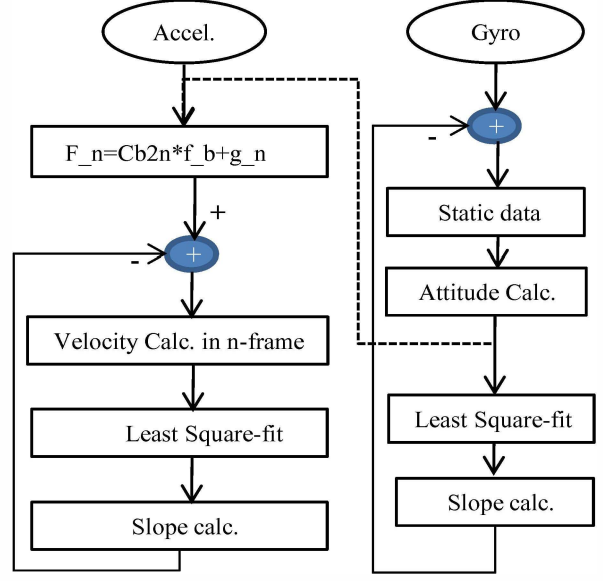


Fig. 1. IMU calibration algorithm using least square fit.

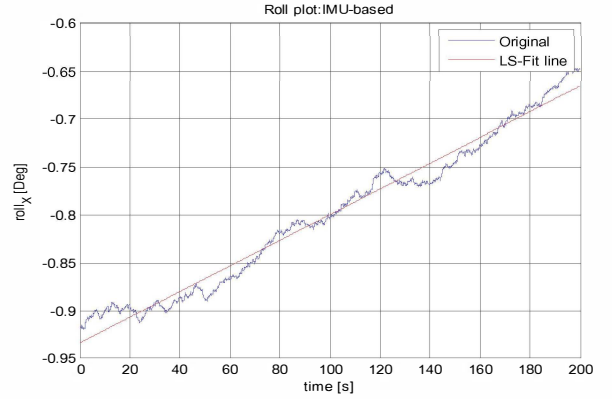


Fig. 2. Roll angle in stationary condition with least square fit.

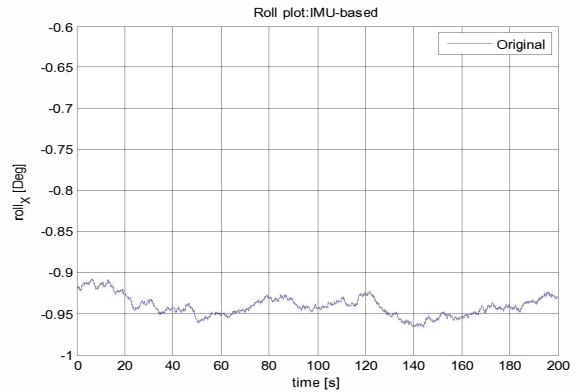


Fig. 3. Roll angle stabilization after compensating with error-factor.

C. Pre-filtering

Another unavoidable factor which must be considered for stabilized attitude estimation is Engine vibration of quadraped robot. This robot is powered by a gasoline engine that drives a hydraulic actuation system, and the

legs are articulated like an animal. It has the compliant elements that absorb shock and recycle the energy from one step to the next [13]. To mitigate the effects of vibration due to different factors, mainly due to engine, we do some pre-filtering on the IMU data as explained below.

The engine vibration has a detrimental affect on the inertial data. Vibration is generally a combination of both angular and translational motion. If bandwidth of both accelerometer and gyro is equal and the vibration of the vehicle at frequencies above this bandwidth has a smaller magnitude than the sensor resolution, in that case the vibration motion can be ignored. However as we are using tactical grade IMU for quadruped attitude estimation, our IMU can detect engine vibration well and we can analyze the IMU data in stationary and moving state in frequency domain in order to assess the engine vibration frequency band. Using real data from quadruped walking robot, we found that:

- Engine vibration is a big issue in interpreting IMU data.
- But fortunately, in our application, we found that engine vibration frequency is well separated from Motion dynamics frequency.
- After performing FFT, we found that engine vibration is above 30Hz, whereas motion dynamics is around (0~20 Hz).

In this work we have designed a 5th order Butterworth low-pass filter applied to IMU data in order to stop engine vibrations to creep into IMU data to disturb motion dynamics estimation.

D. Background Theory

Filters are classified according to the functions that they are to perform, in terms of ranges of frequencies. We will be using low-pass filter here. The highest frequency to be transmitted is f_c , which is also called the cutoff frequency. Frequencies above cutoff are prevented from passing through the filter and they constitute the filter stopband. We used f_c as 25(Hz) and 5th order filter for our purpose.

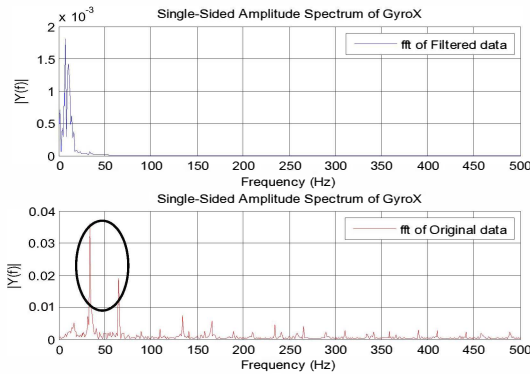


Fig. 4a. FFT of gyro data: filtered (blue) and original (red) data

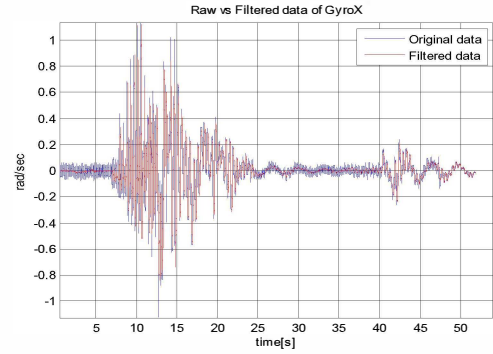


Fig. 4b. Gyro data in time domain after filtering

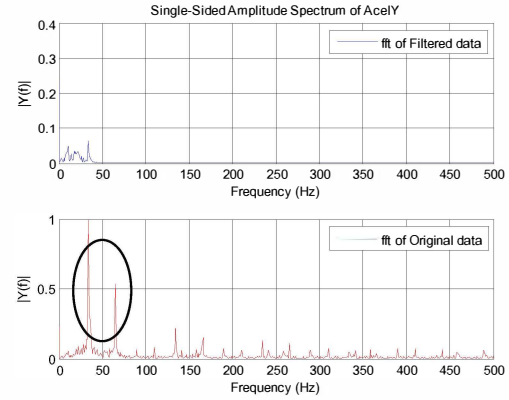


Fig. 5a. FFT of Accel. data: filtered(blue) and original(red) data.

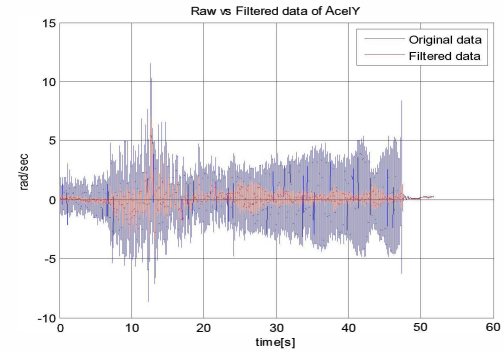


Fig. 5b. Acceleration data in time domain after filtering

After applying the pre-filtering to IMU data, it becomes free of unwanted vibration effects, Fig.4,5. Care must be taken in choosing the cut-off frequency which is pre determined by analyzing the raw IMU data in frequency domain.

E. Attitude Estimator Design

For estimating the Quadruped robot attitude, we have adopted Extended Kalman filter being well known and simple algorithm used extensively in practical nonlinear estimation problems. For Kalman filters, we have two stages of estimation: prediction stage and update stage. Prediction stage uses system dynamics to predict the estimate of desired states along with uncertainty measure in the form of error covariance matrices. While update step makes use of the actual measurement streams which is related with states variables in somehow, linearly or non-linearly, and calculate a weighting matrices called

Kalman gain. In state update, Kalman gain plays a vital role in weighing between the predicted measurement and actual arriving measurements. Thus recursively, it updates the estimates in real time. System dynamics model for use in EKF has already been derived in Section II-A. Here we will present the observation model for use in EKF algorithm.

F. Observation Model

Acceleration sensed by accelerometer in static state can be represented by mathematical form as:

$$\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = g \begin{bmatrix} -\sin \theta \\ \cos \theta \sin \phi \\ \cos \theta \cos \phi \end{bmatrix} \quad (3)$$

where g is gravity and θ and ϕ are pitch and roll angles respectively. From (3) we can derive for pitch and roll angles in terms of sensed acceleration i.e.

$$\mathbf{z}_k(\phi, \theta) = \begin{bmatrix} \tan^{-1}\left(\frac{a_y}{a_z}\right) \\ \sin^{-1}\left(-\frac{a_x}{g}\right) \end{bmatrix} \quad (4)$$

which is represented as measurement model.

$$\mathbf{z} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{Bmatrix} \phi \\ \theta \\ \varphi \end{Bmatrix} + \mathbf{v} \quad (5)$$

$$\mathbf{z} = \mathbf{H}\mathbf{x} + \mathbf{v}$$

Here $\mathbf{v}$ is noise of the measurement sensors.

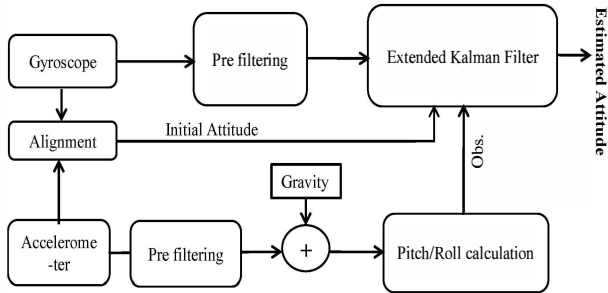


Fig. 6. Attitude estimation schematic diagram.

III. EXPERIMENT METHODOLOGY

Quadruped robot named JinPoong used in this work, is developed by Korea Institute of Industrial Technology (Kitech). A CAD diagram of JinPoong is given in Fig. 7. This robot can be operated in two modes: with power pack (with battery power) and with engine power. Gasoline engine can operate with minimum and maximum RPM as per power requirements of the walking robot.

A. JinPoong Quadruped Robot Description

A brief description of quadruped walking platform used in this work is given here. It is composed of four legs, an engine, power package and hydraulic rotary actuators to carry heavy loads and to move fast on uneven terrain. The size of this robot is 1.2m x 1m x 0.4m (height, length and width, respectively) and the degree of freedom is fifteen. It can carry over 100kg total weight including 60kg payload. The overall control system consists of a main controller system, a hydraulic servo system and an engine/hydraulic power control system [13].

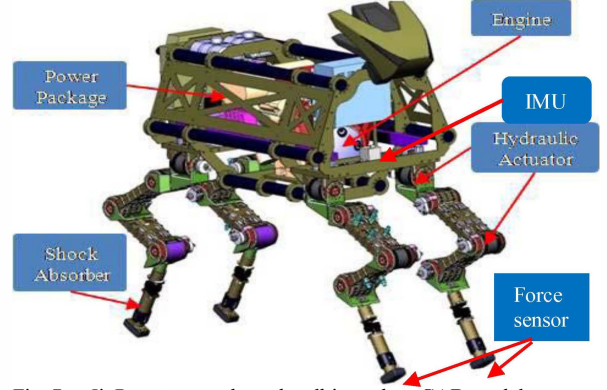


Fig. 7. JinPoong, a quadruped walking robot: CAD model.

The robot was operated in two modes, power pack and engine power, in order to check the effect of engine vibration on calculated attitude. In power pack mode, engine is not started and power is provided externally to operate the robot. In second mode, when engine is turned “ON”, its vibration has pronounced effect on attitude if not pre-filtered, as shown in Fig. 8a and 9a. So we have to apply pre-filtering to suppress unwanted engine vibration not to creep into IMU data and affect robot attitude. The “cleaned” attitude is shown in Fig. 8b and 9b.

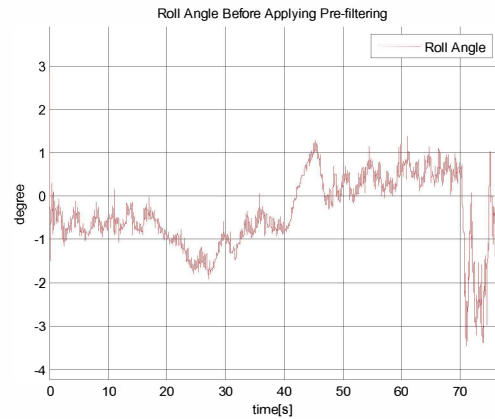


Fig. 8a. Engine-power mode: Roll angle before applying pre-filtering

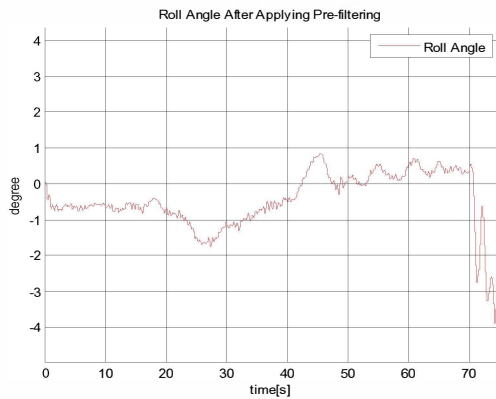


Fig. 8b. Roll angle after applying pre-filtering

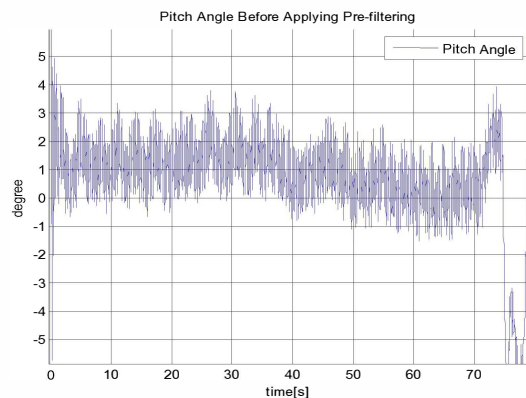


Fig. 9a. Engine-power mode: Pitch angle before applying pre-filtering

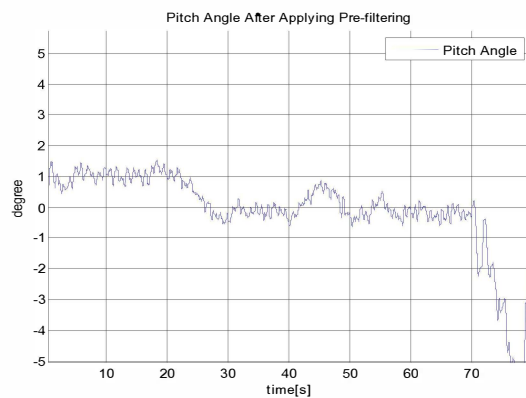


Fig. 9b. Pitch angle after applying pre-filtering

IV. CONCLUSIONS AND FUTURE WORK

At the end, we conclude that stabilized attitude estimation is very important for stable working of quadruped robot, especially in vertical plane the roll and pitch accuracy determines the stability of robot. For estimating the stabilized attitude we must pay attention to certain critical factors. Among them we have addressed the IMU data calibration and pre-filtering of engine vibration in this work. Without performing these tasks, the estimated attitude would be completely unstable and can never stabilize the walking robot. However, by applying proper calibration and pre-filtering we see that estimated attitude is quite clean and stable.

Future experiments will focus on achieving stabilized

attitude for longer time by multi-sensor fusion methods.

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